
I wear a dress
of pink,
white, or red


Vinka flower
Catharanthus Rose

CATHARNTUS ROSE

Scientific Name: Catharanthus roseus (L.) G.Don
Common Names: Madagascar Periwinkle, Rosy Periwinkle, Bright Eyes
Plant Family: Apocynaceae (Dogbane family)
Morphological Characteristics:
Flowers: Salverform 5-petaled blossoms ranging from vibrant pink and purple to snowy white with a distinct darker central eye.
Foliage: Glossy, dark green, oval leaves with a prominent light-colored midrib.
Growth Habit: Evergreen subshrub reaching up to 1 meter in height, blooming continuously in warm climates.



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01. WHEN

First documented by European botanists in the mid-18th century (1750s) after seeds were brought from Madagascar to Paris and London, the plant received its first formal classification in 1759 when Carl Linnaeus named it Vinca rosea. It retained this name until 1837, when Scottish botanist George Don recognized distinct floral structures and officially reclassified it under the new genus Catharanthus.

02. WHERE

Catharanthus roseus is native exclusively to the tropical island of Madagascar, off the southeastern coast of Africa in the Indian Ocean. In its wild native state, the plant evolved as an essential part of Madagascar’s unique coastal ecosystem, thriving in dry sandy dunes, limestone outcrops, forest edges, and arid grassland biomes.Beginning in the mid-18th century, European botanical expeditions transported seeds and live specimens from Madagascar to major botanical hubs in Europe, such as the Jardin des Plantes in Paris and the Chelsea Physic Garden in London



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Ancient Uses



01

Traditional Antidiabetic :
Leaf decoctions were used in Madagascar, India, and the Caribbean to manage blood glucose levels.

02

Cardiovascular Support:
Root extracts served as sedatives to manage high blood pressure (hypertension).

03

Topical Antiseptic & Wound Healing :
Poultices and leaf pastes are applied to skin lesions, minor wounds, insect bites, and oral sores.

04

Fever & Malaria Relief:
Used in traditional island herbalism to lower body temperature during malarial fevers.

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Modern Uses

01

Childhood Leukemia Treatment

A key chemotherapy medication used to treat acute lymphoblastic leukemia (ALL) and Wilms' tumor (kidney tumors) in children.

02

Lymphoma & Cancer Therapy

An essential chemotherapy agent used in combination therapies for Hodgkin's lymphoma, advanced testicular cancer, and non-small cell lung cancer.

03

Lowering High Blood Pressure

Extracted from roots, it relaxes blood vessel walls and lowers vascular resistance to treat hypertension.

04

Improving Brain Blood Circulation

Relaxes cranial blood vessels to boost oxygen and blood flow to the brain, supporting cognitive health and addressing age-related cerebral vascular insufficiency.

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Genetic Engneering

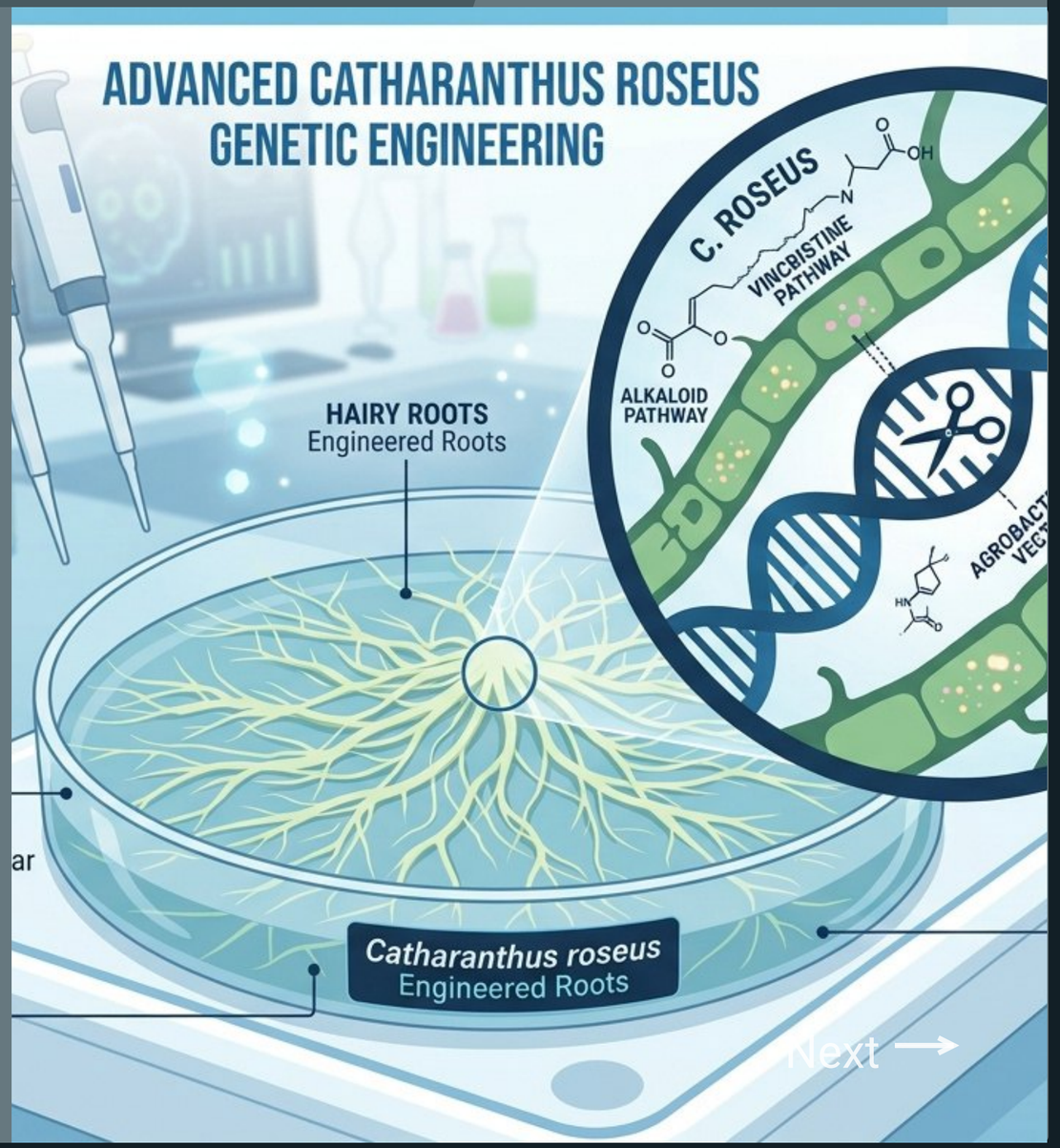
1. Microbial Yeast Factories:
Instead of waiting months to grow plants, scientists transferred the entire genetic "factory code" into yeast cells, allowing us to brew life-saving cancer medications directly in lab fermenters!

2. Magical "Hairy Roots":
Using bacterial gene transfer, plant roots are stimulated to grow at lightning speed in sterile labs—acting as continuous, year-round bioreactors to produce blood pressure and circulation compounds without depending on seasons or soil.

3. Smart Genetic Switches:
Rather than tweaking just one gene, scientists now control "master genes" (like ORCA3) to unlock the plant's full production lines, multiplying its natural drug output several times over.

4. Cellular Smart Pumps:
To prevent the plant cells from poisoning themselves with too much internal medicine, researchers engineered protein "pumps" on cell walls to flush the medicinal compounds out instantly, boosting overall yields.

5. Precision Gene Editing (CRISPR/Cas9):
Using state-of-the-art genetic editing, scientists shut down unnecessary side pathways, steering 100% of the plant's energy and resources into mass-



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Our team

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